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United States Patent	12398966
Kind Code	B2
Date of Patent	August 26, 2025
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Backward compatible safety slide firing pin-striker system

Abstract

A firearm system which allows firearm disassembly without trigger actuation. The system includes a slide having a divot or cavity disposed laterally off of the main channel. This divot allows for the pivoting of the striker sleeve such that the striker tang sear can move between a first position that is engaged with a corresponding trigger bar sear of a frame and a second position whereby the striker tang sear is disposed at least partially within the divot. This allows the striker tang sear to clear the trigger bar sear for disassembly without requiring a pull of the trigger.

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Appl. No.:	19/017927
Filed:	January 13, 2025

Prior Publication Data

Document Identifier	Publication Date
US 20250237450 A1	Jul. 24, 2025

Related U.S. Application Data

us-provisional-application US 63624297 20240124

Publication Classification

Int. Cl.: F41A11/00 (20060101); F41A17/02 (20060101); F41A17/56 (20060101); F41A17/64 (20060101)

U.S. Cl.:

CPC **F41A11/00** (20130101); **F41A17/02** (20130101); **F41A17/56** (20130101); **F41A17/64** (20130101);

Field of Classification Search

CPC: F41A (17/02); F41A (17/56); F41A (17/64); F41A (11/00)

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Background/Summary

(1) This application claims priority to U.S. provisional application 63/624,297, filed Jan. 24, 2024. U.S. provisional application 63/624,297 and all other extrinsic references contained herein are incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The field of the invention is firearms and firearm safety.

BACKGROUND

(2) The background description includes information that may be useful in understanding the

present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

(3) A popular firearm design requires trigger actuation for field stripping/disassembly. This could result in unintended discharge if the firearm was not properly cleared of live ammunition before disassembly. An unintended discharge of a loaded firearm could cause serious injury or death.

(4) Thus, there is still a need for a safe way to disassemble a firearm that doesn't require a trigger pull.

SUMMARY OF THE INVENTION

(5) The inventive subject matter provides apparatus, systems and methods in which a slide for a firearm includes a divot disposed laterally from its central channel and aligned such that a striker tang sear which is coupled to a striker sleeve disposed within the channel can move from a first position (engaged with a trigger bar sear of the firearm frame) to a second position at least partially within the divot.

(6) When the striker tang sear is at the second position, it clears the trigger bar sear when the slide is moved forward along the length of the firearm, allowing for easy removal of the slide without requiring trigger activation.

(7) The inventive subject matter further includes a backplate that includes a rotatable lock. The rotatable lock engages with the striker sleeve such that the rotation of the lock causes a corresponding rotation of the striker sleeve (and thus, the movement of the coupled striker tang sear between the first and second positions).

(8) The rotatable lock can include an actuation element that can be used by the user to rotate the rotatable lock. The actuation element can be a groove, a shaped cavity, a textured surface, an externally-extending tab(s), or other type of user friendly form that allows for easy rotation of the lock.

(9) Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

(10) All publications identified herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

(11) The following description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

(12) In some embodiments, the numbers expressing quantities of ingredients, properties such as concentration, reaction conditions, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term "about."

Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

(13) Unless the context dictates the contrary, all ranges set forth herein should be interpreted as

being inclusive of their endpoints and open-ended ranges should be interpreted to include only commercially practical values. Similarly, all lists of values should be considered as inclusive of intermediate values unless the context indicates the contrary.

(14) As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(15) The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

(16) Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) FIG. 1 is a left side (partial cutaway, muzzle toward left side of page) view of a Prior Art Slide Firing Pin/Striker System of a popular type of firearm in a Pre Disassembly Initiated State.

(2) FIG. 2 is a rear-left-bottom isometric view of FIG. 1.

(3) FIG. 3 is a detailed view of Section A of FIG. 2.

(4) FIG. 4 is a left side view of the Prior Art Slide Firing Pin/Striker System of FIG. 1 in a Disassembly Initiated State showing the trigger partially pulled.

(5) FIG. 5 is a left side view of the Prior Art Slide Firing Pin/Striker System of FIG. 1 with the trigger fully pulled.

(6) FIG. 6 is a left side view of the Prior Art Slide Firing Pin/Striker System of FIG. 1 in a Mid Disassembly State.

(7) FIG. 7 is a left side view of the Prior Art Slide Firing Pin/Striker System of FIG. 1 in a Fully Disassembled State.

(8) FIG. 8 is a rear-left-bottom exploded isometric view of the components of a Firing Pin/Striker Assembly.

(9) FIG. 9 is a left side view of FIG. 8.

(10) FIG. 10 is a rear-left-bottom exploded isometric view of the components of FIG. 8 after a first step of reassembly.

(11) FIG. 11 is a rear-left-bottom exploded isometric view of the components of FIG. 8 after a subsequent step of reassembly from the state of FIG. 10.

(12) FIG. 12 is a rear-left-bottom exploded isometric view of the components of FIG. 8 after a subsequent step of reassembly from the state of FIG. 11.

(13) FIG. 13 is a rear-left-bottom exploded isometric view of the components of FIG. 8 after a

subsequent step of reassembly from the state of FIG. 12.

(14) FIG. 14 is a rear-left-bottom isometric view of a fully assembled Firing Pin/Striker Assembly (of the components of FIG. 8) after a final step of reassembly from the state of FIG. 13.

(15) FIG. 15 is a left side view of FIG. 14.

(16) FIG. 16 is a rear-left-bottom exploded isometric view of the components of a Rotatable Lock and Backplate Assembly, according to embodiments of the inventive subject matter.

(17) FIG. 17 is a detailed view of Section B of FIG. 16.

(18) FIG. 18 is a left side view of FIG. 16.

(19) FIG. 19 is a front-left-top isometric view of FIG. 16.

(20) FIG. 20 is a rear-left-bottom isometric view of a Backward Compatible Safety Slide Firing Pin/Striker System in a Pre Disassembly Initiated State, according to embodiments of the inventive subject matter.

(21) FIG. 21 is a detailed view of Section C of FIG. 20.

(22) FIG. 22 is a rear-left-bottom isometric view of a Backward Compatible Safety Slide in isolation, according to embodiments of the inventive subject matter.

(23) FIG. 23 is a left side view of FIG. 20.

(24) FIG. 24 is a left side view of the Backward Compatible Safety Slide Firing Pin/Striker System of FIG. 20 when an undesirable unlocking is attempted, according to embodiments of the inventive subject matter.

(25) FIG. 25 is a rear side view of FIG. 24.

(26) FIG. 26 is a detailed view of Section D of FIG. 25.

(27) FIG. 27 is a rear-left-bottom isometric view of the Backward Compatible Safety Slide Firing Pin/Striker System of FIG. 20 in a Disassembly Initiated State.

(28) FIG. 28 is a detailed view of Section E of FIG. 27.

(29) FIG. 29 is a left side view of FIG. 27.

DETAILED DESCRIPTION

(30) The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

(31) As used herein, and unless the context dictates otherwise, the term “coupled to” is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms “coupled to” and “coupled with” are used synonymously.

(32) FIG. 1 is a left side view of a Prior Art Slide Firing Pin/Striker System **100** (hereinafter “System **100**”) of a popular type of firearm in a “Pre Disassembly Initiated State”. The Pre Disassembly Initiated State can be considered to be the state of normal use of the firearm. This is prior to taking any steps necessary for the disassembly of the weapon.

(33) System **100** can be considered to comprise a Prior Art Slide **101** (hereinafter “Slide **101**”) which includes a Firing Pin/Striker Assembly **150** and a Prior Art Backplate **105**.

(34) A Striker **102**, two Half Clam Spring Capture Cups **122**, Spring **121** and Striker Sleeve **104** make up the Firing Pin/Striker Assembly **150**.

(35) Slide **101** further comprises Striker Channel **115** and Prior Art Slide Rail Grooves **120** (visible in FIG. 3).

(36) Striker **102** further comprises a Striker Tang Sear **103** (shown “cheese-holed” for illustrative clarity).

(37) System **100** is functionally coupled with a Frame **106** of the firearm. Frame **106** further comprises Trigger **107**, trigger bar Sear **108**, Front Frame Rails **119** and Aft Frame Rails **118**.

(38) Frame **106** is illustrated as 2 separate pieces (for illustrative ease only). In practice, the small upside down pyramid is connected to the main body of the Frame **106**.

(39) During normal operations, the actuation of Trigger **107** causes trigger bar Sear **108** to compress Spring **121** (thus imparting potential energy into the Spring **121**). Additional/further actuation of Trigger **107** causes trigger bar Sear **108** to release Striker **102** which allows the Spring's potential energy to convert to kinetic energy. Kinetic energy of the released Striker **102** then discharges ammunition chambered in the firearm. Subsequent force of discharging gunpowder gases initiates a cycling process which (in a known manner) ejects the spent ammunition case, chambers live ammunition, and resets the system for subsequent Trigger **107** actuation.

(40) FIG. **2** is a rear-left-bottom isometric view of FIG. **1**.

(41) FIG. **3** is a detailed view of Section A of FIG. **2**. It is important to note that System **100** is fixed in the Y Axis (i.e.—only moves along the length of the firearm and not allowed to displace up/down) from Frame **106** due to the communication between Frame Rails (**118/119**) and Prior Art Slide Rail Grooves **120**.

(42) FIG. **4** is a left side view of System **100** of FIG. **1** in a Disassembly Initiated State showing the Trigger **107** partially pulled/actuated. Actuation of Trigger **107** causes trigger bar Sear **108** to displace to the rear (+X Direction) and down (−Y Direction) in a known manner. Top of the trigger bar Sear **108** is at the very bottom of Striker Tang Sear **103** and any further actuation of the Trigger **107** will cause trigger bar Sear **108** to displace further down (in the −Y Direction) and release the Striker **102**.

(43) FIG. **5** is a left side view of System **100** of FIG. **1** with the Trigger **107** fully pulled. The fully displaced (rearward and downward) trigger bar Sear **108** releases Striker **102** (under the biasing force of Spring **121**) and creates a Disassembly Clearance **109** between Striker Tang Sear **103** and trigger bar Sear **108**. This action completes the first step of the Disassembly Process.

(44) Only after Disassembly Clearance **109** is created (which allows Striker Tang Sear **103** to translate horizontally over and clear the trigger bar Sear **108**) can the firearm be further disassembled. Additional steps and components that must be actuated to unlock Slide **101** and allow it to move forward off Frame **106** are well known and thus not described or illustrated.

(45) FIG. **6** illustrates how the Slide **101** has been pushed forward along the length of the firearm (−X Direction) in what could be considered a Mid Disassembly State. At this point in the disassembly process, the Striker Tang Sear **103** has cleared the trigger bar Sear **108** and is forward of same.

(46) It is important to reiterate/emphasize that Trigger **107** must be actuated/pulled to create the Disassembly Clearance **109** (which in turn allows further disassembly). As is commonly known and discussed above, actuating the Trigger **107** is also how to fire/discharge the firearm. Thus, the reader will appreciate that requiring the pressing of the Trigger **107** to initiate the disassembly of the firearm carries significant risks. If live ammunition is still chambered in the firearm, a “Negligent/Accidental” discharge of the firearm would occur (which could cause injury or death to the user or others).

(47) It is also worthwhile to note that there are currently more than 50 million firearms utilizing System **100** of FIG. **1**. Negligent/Accidental discharges of this type of firearm occur on a weekly (if not almost daily) basis.

(48) FIG. **7** is a left side view of System **100** of FIGS. **1-3** in a Fully Disassembled State where Slide **101** has been displaced forward enough to clear both Front and Aft Frame Rails (**119/118**). At this point, Slide **101** can be separated from the Frame **106** of the firearm for cleaning, repairs, storage, etc.

(49) FIG. **8** is a rear-left-bottom exploded isometric view of the components of a Firing Pin/Striker Assembly **150**. The Firing Pin/Striker Assembly **150** includes a Striker **102**, two Half Clam Spring Capture Cups **122**, Striker Sleeve **104** and Spring **121**.

(50) Striker **102** further comprises a Striker Tang Sear **103** and Half Clam Spring Capture Cup

Edge **123**.

(51) Striker Sleeve **104** further comprises a Striker Rotation Edge **112**, Bridge **113** and Keyed Lock Interface **114**.

(52) FIG. **9** is a left side view of FIG. **8**.

(53) FIGS. **10** through **14** show the step-by-step assembly process of the components of the Striker Assembly **150**. Striker Sleeve **104** is slid onto Striker **102** (FIG. **10**). Spring **121** is then slid over Striker **102** and compressed (FIG. **11**). Both Half Clam Spring Capture Cups **122** are placed around the central shaft of the Striker **102** near its Half Clam Spring Capture Cup Edge **123** (FIGS. **12/13**). Spring **121** is then released to capture the two Half Clam Spring Capture Cups **122** against the Half Clam Spring Capture Cup Edge **123** (FIGS. **14/15**).

(54) FIG. **15** is a left side view of the fully assembled Striker Assembly **150**.

(55) It is important to note that the prior art was configured to allow only transitional (i.e.—fore/aft along the X axis) movement of the components of the Striker Assembly **150**. In addition to the transitional movement (of the prior art), the inventive subject matter allows for rotational (i.e.—rotating/twisting about the central axis) movement of the Striker Assembly **150**, as will be described in further detail herein.

(56) It should be easy to envisage that rotating/twisting of Striker Sleeve **104** about its axis would likewise rotate/twist Striker Tang Sear **103** due to its interplay with Striker Rotation Edge **112**.

(57) FIGS. **16** through **19** are various exploded isometric views of the components of a Rotatable Lock and Backplate Assembly **224**, according to embodiments of the inventive subject matter. A Rotatable Lock **214** and Backplate **213** comprise the components of the Rotatable Lock and Backplate Assembly **224**.

(58) Rotatable Lock **214** further comprises three (situated at 120 degrees intervals around its body) Locking Peaks **216**, three Thumb Interface Nubs **222**, Keyed Sleeve Interface **218** and Unlocking Prevention/Displacement Limiter **220**.

(59) Backplate **213** further comprises three (situated at 120 degrees intervals) Locking Peak Divots **215** around its central hollow cylinder and Displacement Limiter Interface **223**.

(60) When assembled in the “Locked/Operational” State, the three Locking Peaks **216** communicate with the three Locking Peak Divots **215**. Keyed Sleeve Interface **218** communicates with Keyed Lock Interface **114** (of Striker Sleeve **104**) and rotationally locks Rotatable Lock **214** to Striker Assembly **150**.

(61) When Striker Assembly **150** is assembled into either (prior art or inventive subject matter) system, the systems are configured to preload the Spring **121** (in a known manner and for a known purpose) so as to affect a gap between the front edge of Striker Tang Sear **103** and Bridge **113** (of Striker Sleeve **104**) as illustrated in FIGS. **1** and **2**. This preloading/compressing of Spring **121** transfers the spring force to bias the three Locking Peaks **216** (of Rotatable Lock **214**) into the “valleys” of the three Locking Peak Divots **215** thus locking the Rotatable Lock **214** and Striker Assembly **150** into the Locked/Operational State.

(62) FIGS. **20**, **21** and **23** correspond to the Pre Disassembly Initiated State of a Backward Compatible Safety Slide Firing Pin/Striker System **200** (hereinafter “System **200**”), according to embodiments of the inventive subject matter.

(63) As with the prior art depictions of FIGS. **1-3**, the state of FIGS. **20**, **21** and **23** would also correspond to the state of normal use/operation of the firearm—to discharge ammunition via actuation of the trigger.

(64) FIG. **20** is a rear-left-bottom isometric view of System **200**, comprising a Backward Compatible Safety Slide **210** (hereinafter “Slide **210**”), Striker Assembly **150**, and Rotatable Lock and Backplate Assembly **224**.

(65) FIG. **22** is a rear-left-bottom isometric view of Slide **210** from FIGS. **20-21**, shown in isolation, according to embodiments of the inventive subject matter. Slide **210** includes Striker Tang Sear Divot **211** (hereinafter “Divot **211**”), Backward Compatible Safety Slide Striker Channel

217 (hereinafter “Channel **217**”) and Backward Compatible Safety Slide Rail Grooves **221** (hereinafter “Grooves **221**”).

(66) As seen in FIGS. **20-23**, Channel **217** is dimensioned to receive Striker Assembly **150** for normal operation of the firearm and also for the disassembly procedure of the inventive subject matter discussed in greater detail herein. Divot **211** is dimensioned to at least partially receive Striker Tang Sear **103** when the Striker Assembly **150** within Channel **217** is rotated as is discussed in further detail herein.

(67) Grooves **221** serve to mate with the Frame Rails **118** and **119** of the Frame **106** to secure the System **200** and allow the appropriate movements during normal operation of the firearm and for assembly/disassembly as discussed herein.

(68) FIG. **21** is a detailed view of Section C of FIG. **20** showing the positional relationships of trigger bar Sear **108**, Striker Tang Sear **103**, Striker Sleeve **104**, Striker Rotation Edge **112**, Divot **211**, Rotatable Lock **214** and Backplate **213**.

(69) FIG. **23** is a left side view of FIG. **20**.

(70) FIGS. **20**, **21** and **23** also illustrate the communication between Striker Assembly **150** and Rotatable Lock and Backplate Assembly **224** in the Locked/Operational State. Striker Tang Sear **103** is in contact with the trigger bar Sear **108** of Frame **106**. Thus, actuation of the Trigger **107** would cause trigger bar Sear **108** to interact with the Striker Tang Sear **103**. If ammunition was in the firearm with a round chambered, this would cause the firearm to discharge.

(71) An important note considering the Unlocking Prevention/Displacement Limiter **220** (FIGS. **17** and **19**) is addressed herein.

(72) If not for the Unlocking Prevention/Displacement Limiter **220** (FIGS. **17** and **19**), System **200** could transition to an “Unlocked/Disassembly” State (rendering the firearm inert) should the firearm be roughly handled (e.g.—when an assailant attempts to wrestle away a police officer's firearm). In other words, due to the Limiter's design, no external force can cause System **200** to transition to the Unlocked/Disassembly State from the Locked/Operational State other than actuation via the rotatable lock **214**.

(73) FIG. **24** is a left side view of System **200** when an undesirable unlocking is attempted, according to embodiments of the inventive subject matter.

(74) FIG. **25** is a rear side view of FIG. **24**.

(75) As illustrated in the detailed view of FIG. **17**, the Unlocking Prevention/Displacement Limiter **220** is configured like an arm that sticks out of the piston-like body of the Rotatable Lock **214**. Simply put, the end of the arm is configured to interfere with Frame **106** to prevent unlocking from the Locked/Operational State.

(76) FIG. **26** is a detailed view of Section D of FIG. **25** illustrating the aforementioned interference of the Unlocking Prevention/Displacement Limiter **220** against the Frame (illustrated here as an upside down pyramid) **106**. In preferred embodiments, this interference occurs at 2 degrees or less out-of-vertical for the rotationally locked components of Striker Tang Sear **103** and Rotatable Lock **214**. The tip of Unlocking Prevention/Displacement Limiter **220** is shown abutted against the base of upside down pyramid portion of Frame **106** and Striker Tang Sear **103** is approximately 2 degrees out of vertical (but still in communication with trigger bar Sear **108**).

(77) FIG. **27** is a rear-left-bottom isometric view of the System **200** of FIGS. **20-23**, in the Disassembly Initiated State, according to embodiments of the inventive subject matter. The first step in the disassembly process is to manipulate Slide **210** to its rearmost travel limit and lock it in the “Open Action” State.

(78) The Open Action State is a well known state where the slide of a firearm is locked back/open (by other known operations and components) to unload, clear malfunctions, or facilitate reloading.

(79) The Open Action State displaces the Unlocking Prevention/Displacement Limiter **220** (of the Rotatable Lock **214**) away from the Frame **106** (to preclude Limiter interference). As a point of clarification/reiteration, System **200** can only be Unlocked or Locked in the Open Action State.

(80) System **200** is transitioned to the Unlocked/Disassembly State (with reference to FIGS. **21** and **17**) by pushing/displacing Rotatable Lock **214** (via the Thumb Interface Nubs **222** and against the biasing force of the Spring **121** of the Striker Assembly **150**) sufficiently enough so that its three Locking Peaks **216** (which were previously communicating with the “valleys” of the three Locking Peak Divots **215**) clears the “hills” of the three Locking Peak Divots **215** as Rotatable Lock **214** is subsequently rotated counterclockwise (in the shown example, approximately 60 degrees). This rotation aligns its three Locking Peak **216** bodies with the correspondingly configured semi circular grooves (formed between the three Locking Peak Divots **215**) within the central hollow cylindrical opening of Backplate **213**.

(81) The rotation/twisting of the Rotatable Lock **214** causes the rotationally locked Striker Tang Sear **103** to rotate/“tuck” wholly or partially into the properly configured and aligned Divot **211** of Slide **210**.

(82) The amount that Striker Tang Sear **103** moves into Divot **211** can be dictated by factors such as the size of the Striker Tang Sear **103** and the relative width of Slide **210**. However, it should be understood that the movement of Striker Tang Sear **103** into Divot **211** is sufficient for the Striker Tang Sear **103** to clear the trigger bar Sear **108** as discussed herein.

(83) As illustrated in the detailed view of FIG. **28**, subsequently releasing thumb pressure from the three Thumb Interface Nubs **222** allows force of Spring **121** to displace the Striker Sleeve **104** and Rotatable Lock **214** rearward to remove the previously described (at the end of descriptions of FIGS. **16-19**) gap between the front edge of Striker Tang Sear **103** and Bridge **113** (of Striker Sleeve **104**) while simultaneously abutting the appropriately configured Unlocking Prevention/Displacement Limiter **220** (of Rotatable Lock **214**) against Displacement Limiter Interface **223** (of Backplate **213**). Interference between Unlocking Prevention/Displacement Limiter **220** and Displacement Limiter Interface **223** contains Striker Assembly **150** within Slide **210**.

(84) Displacement of Rotatable Lock **214** into Backplate **213** also slides the bodies of the three Locking Peaks **216** (referencing FIG. **17**) into the correspondingly configured semi circular grooves (formed between the three Locking Peak Divots **215**) within the central hollow cylindrical opening of the Backplate **213**. This rotationally locks the Striker Assembly **150** (and thus its Striker Tang Sear **103**) in the “tucked” position in Divot **211**.

(85) Rotatable Lock **214** will stand proud of Backplate **213** at the end of this action to visually inform the user that System **200** is now in the Unlocked/Disassembly State.

(86) FIG. **29** is a left side view of FIG. **27**. Tucking the Striker Tang Sear **103** into Divot **211** creates a Rotationally Generated Disassembly Clearance **209** between Striker Tang Sear **103** and trigger bar Sear **108**.

(87) The Rotationally Generated Disassembly Clearance **209** allows the Striker Tang Sear **103** to translate horizontally over and clear the trigger bar Sear **108** to allow further disassembly without Trigger **107** actuation (as was required by the prior art). This difference is clearly illustrated in the different positions of Triggers **107** in System **100** of FIG. **5** and System **200** of FIG. **29**. Additional steps and components that must be actuated to unlock Slide **210** and allow it to move forward off Frame **106** are well known and thus not described or illustrated.

(88) The Rotatable Lock **214** of the described embodiment utilizes three Thumb Interface Nubs **222** as Actuation Elements of System **200**. As implied by the name, the Thumb Interface Nubs **222** interface with the user's thumb or finger.

(89) In other embodiments, an Actuation Element of rotatable lock **214** can be in the form of a groove can be dimensioned to be actuated by a common object such as a coin, fingernail, or a flat-head screwdriver.

(90) In alternative embodiments, the Actuation Element of rotatable lock **214** can be a cavity dimensioned and shaped to receive a specific tool such as a hexagonal shaped-tool, a Phillips-head screwdriver, or a custom-shaped tool.

(91) In yet other embodiments, the Actuation Element of rotatable lock **214** can be a tab extending outward from the Lock that the user can grip between two or more fingers.

(92) It should be appreciated that in addition to the safer disassembly methods discussed herein, the systems of the inventive subject matter serve as a safety even when disassembly is not intended. If Slide **210** is moved to its operational/closed position from the State of FIG. **29**, the firearm cannot be discharged by pulling the Trigger **107**. Pulling the Trigger **107** will actuate the trigger bar Sear **108** but, because the trigger bar Sear **108** is no longer engaged with the Striker Tang Sear **103**, the Striker Assembly **150** will not be actuated.

(93) The system **200** could be incorporated into new firearm designs and is also backward compatible by simply utilizing a Prior Art Backplate **105** seen in FIG. **1** instead of Backplate **213** and Rotatable Lock **214** of the present invention. This allows for the Slide **210** to be used within a conventional firearm in a conventional fashion, should the user choose to disable the safety benefits discussed herein. It should be noted that using a conventional Prior Art Backplate **105** would revert the system to requiring trigger actuation for disassembly.

(94) The invention could be manufactured using any industry standard materials (metals, alloys, sheet metal, plastics, etc.) and processes (injection molding, MIM, sheetmetal folding, additive manufacturing/3D printing, subtractive machining, etc.). Alternative embodiments, combinations, and/or uses of the methods and devices described above and obvious modifications and equivalents thereof are intended to be included within the scope of the invention.

(95) Although the described Backward Compatible Safety Slide Firing Pin/Striker System is for a popular brand of firearm, it is not meant to be limited to just that brand or type (i.e.—the invention is applicable to other brands and/or handguns, rifles etc.).

(96) It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refers to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

Claims

1. A system for safely disassembling a firearm, comprising: a slide having a channel, the channel dimensioned to receive a striker sleeve; the striker sleeve disposed within the channel and dimensioned to receive a striker; the striker having a striker tang sear, wherein the striker tang sear is aligned with a trigger bar sear of a frame of the firearm when the striker tang sear is at a first tang sear position; and wherein the slide includes a divot extending laterally from the channel, the divot disposed on the slide such that the divot becomes aligned with the striker tang sear when the slide is at an open action state, and the divot dimensioned to at least partially receive the striker tang sear when the striker tang sear is moved to a second tang sear position.
2. The system of claim 1, further comprising a backplate, the backplate further comprising a rotatable lock that is coupled with the striker sleeve such that movement of the rotatable lock causes the striker sleeve to rotate between the first tang sear position and the second tang sear position.
3. The system of claim 2, wherein the rotatable lock further comprises an actuation element.
4. The system of claim 3, wherein the actuation element comprises at least one of a groove, a cavity

dimensioned to receive a tool, a textured surface, or an outwardly-extending tab.

5. The system of claim 1, wherein the striker tang sear is not aligned with the trigger bar sear when the striker tang sear is at the second tang sear position.
